MILITARY SPECIFICATION

BENZOYL PEROXIDE GEL

This specification is approved for use by the Defense Personnel Support Center, Defense Logistics Agency, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Benzoyl Peroxide Gel.

1.2 Classification. The Benzoyl Peroxide Gel covered by this specification shall be of the following strengths (see 6.2):

Strength 1 - 5 percent
Strength 2 - 10 percent

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Benzoyl Peroxide in a suitable gel formulation of either alcohol, aqueous or acetone base.

Strength 1 - Shall contain 5 percent Benzoyl Peroxide.
Strength 2 - Shall contain 10 percent Benzoyl Peroxide.

3.1.1 Assay.

3.1.1.1 Strength 1. The finished gel shall assay to contain not less than 90.0 percent and not more than 140.0 percent of the required amount of benzoyl peroxide when determined as specified in 4.7.1.

3.1.1.2 Strength 2. The finished gel shall assay to contain not less than 90.0 percent and not more than 125.0 percent of the required amount of benzoyl peroxide when determined as specified in 4.7.1.

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3.1.2 Identity. The presence of Benzoyl Peroxide in the finished gel shall be determined using the Identificatin method in USP monograph for Benzoyl Peroxide Lotion. Note: Should USP Hydrous Benzoyl Peroxide Reference Standard not be available, as an alternate, a standard benzoyl peroxide which has been tested against the requirements in the monograph for Hydrous Benzoyl Peroxide, USP, and is of comparable purity may be used. Test records delineating the results shall be made available, if requested. Test comparison for each lot of standard with Hydrous Benzoyl Peroxide, USP, need be performed one time only.

3.2 Ingredients.

3.2.1 Hydrous benzoyl peroxide. The hydrous benzoyl peroxide used in the manufacture of the gel shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.3 Other ingredients. All other ingredients used in the manufacture of the gel shall be of USP or NF quality, or if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade suitable for use in the gel.

3.4 Expiration dating period. The expiration dating period shall be not less than 24 months.

3.4.1 Delivery. Not more than 3 months of the expiration dating period shall have elapsed at time of delivery to the Government.

3.4 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiraton dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

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4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of gel produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled tube.
Minor	II	2.5	Filled tube.

4.3.1 End item testing. Noncompliance with any of the following is cause for rejection. The acceptance number for each test shall be zero (0).

TABLE II. Criteria for end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Assay	3.1.1	S-2	5
Identity	3.1.2	USP	

1/ Where possible the sample shall be used for two or more tests.

4.4 Examination. The finished product shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3 and 4.3.1.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Gel not free from foreign matter.
102	Gel not homogeneous.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks and tears.

1/ Inspection is not restricted to the classified possible defects listed above.

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4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests.

4.7.1 Assay for strength 1 and strength 2. The samples shall be assayed as directed in the USP monograph for Benzoyl Peroxide Lotion. As an alternate, the sample may be assayed by the following method.

Reagents.

Acetonitrile, ACS grade
Ethyl Benzoate, Analytical Reference Standard
Hydrous Benzoyl Peroxide, USP Reference Standard
See note in 3.1.2 Identity, concerning USP Hydrous Benzoyl Peroxide Reference Standard.

Standard Preparation

A. Internal Standard Preparation (Ethyl Benzoate).

Weigh 350 mg (± 20 mg) reference standard ethyl benzoate into a 100 ml volumetric flask and make to volume with acetonitrile. Mix well.

B. Standard Preparation (Benzoyl Peroxide).

Weigh 65 mg (± 5 mg) USP reference standard benzoyl peroxide accurate to ± 0.1 mg into a tared 50 ml volumetric flask and keep the flask stoppered to avoid loss of moisture during weighing. Dissolve in and make to volume with acetonitrile. Pipet 10.0 ml of this standard solution and also 5.0 ml of internal standard preparation into a 25 ml volumetric flask. Make to volume with acetonitrile and mix well.

Sample Preparation

Weigh 820 mg (± 50 mg) of Strength 1 sample or 410 mg (± 25 mg) of Strength 2 sample accurate to ± 0.1 mg into a tared 50 ml volumetric flask. Make to volume with acetonitrile; mix well to dislodge the sample sticking to the bottom and sonicate the mixture for 5 minutes Centrifuge the solution at 3400 rpm for about 15 minutes until solution is clear. Pipet 10.0 ml of the clear centrifugate and 5.0 ml of the internal standard preparation into a 25 ml volumetric flask. Make to volume with acetonitrile and mix well.

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Instrumentation

Liquid Chromatograph: Micromeritics Model 7000 or equivalent

Column: 0.26 x 50 cm, stainless steel
Packing: Reverse Phase C_{18} Bondapak/Corasil (Waters Assoc.)
Eluent: 45/55; acetonitrile/purified water (percent v/v) nominally
Flow Rate: 1 ml/min (pressure≈800 psi)
Temperature: ambient

Detector: UV, 254 nm

Optical Density: 0.2 AUFS

Recorder:

Voltage Full Scale: 10 mv
Chart Speed: .5 in/min

NOTE: Other instruments and parameters yielding comparable results may be substituted.

Procedure

Inject 8 µl of the standard preparation in to high pressure liquid chromatograph. Quantitate the peak and calculate the response ratio which is the ratio of the peak quantitation of Benzoyl Peroxide to peak quantitation of ethyl benzoate. Repeat the 8 µl injection at least two more times. Average the response ratios to obtain Qstd. The response ratios for the injections shall agree within 1 percent from their average.

Inject 8 µl of the sample preparation and determine the response ratio. Repeat the 8 µl injection at least two more times. Average the response ratios to obtain Qspl. The response ratios for the injections shall agree within 1 percent from their average.

NOTE: Injection size may vary depending upon instrument and parameters used, in order to obtain adequate quantitation values.

Calculation

$$\text{Percent Benzoyl Peroxide} = \frac{Qspl}{Qstd} \times \frac{\text{mg Std}}{\text{mg Spl}} \times A$$

Where Qstd is the average response ratio for standard preparation
Qspl is the average response ratio for Sample Preparation
A is the percent purity of the Benzoyl Peroxide reference standard.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One tube containing 1.5 oz as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available tube considered acceptable by the Food and Drug Administration Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve (12) units of issue of strength 1 or 2 shall be packaged as specified in 5.2.13 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Two hundred eighty-eight (288) units of issue of strength 1 or strength 2 shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Two hundred eighty-eight (288) units of issue of strength 1 or strength 2 shall be packed as specified in 5.4.3.5.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug, and Cosmetic Act and shall include the information required below:

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5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type II shelf life markings as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C (59° - 86°F)".

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type II shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C (59° - 86°F)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following items:

National Stock Number	Item Identification	Specification Designation
6505-01-036-8628	BENZOYL PEROXIDE GEL, 10%, 1.5 oz	Strength 2
6505-01-036-8629	BENZOYL PEROXIDE GEL, 5%, 1.5 oz	Strength 1

Preparing activity:
DLA-DM

Project No. 6505-1943

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-B-37799(DLA-DM) BENZOYL PEROXIDE GEL

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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